

Cybersecurity Tools Lab Experiment

Experiment: Registry Analysis and Boot-Time Logging Using Process Monitor

1. Aim

To perform **Windows Registry analysis** and capture **boot-time system activity** using **Microsoft Sysinternals Process Monitor (Procmon)**.

2. Objectives

- To understand how Windows processes interact with the Registry.
- To monitor Registry operations such as RegOpenKey, RegQueryValue, RegSetValue, etc.
- To capture system activity during Windows startup.
- To identify processes and services involved in boot-time Registry activity.
- To analyze suspicious or failed Registry operations.

3. Requirements

- Windows 10/11 computer
- Administrator privileges
- Internet connection for downloading Procmon
- **Microsoft Sysinternals Process Monitor**

[Download Process Monitor from Microsoft Sysinternals](#)

4. Theory

The **Windows Registry** is a hierarchical database used by Windows and applications to store configuration information.

Important Registry hives include:

Hive	Purpose
HKEY_LOCAL_MACHINE (HKLM)	System-wide configuration
HKEY_CURRENT_USER (HKCU)	Current user's configuration
HKEY_CLASSES_ROOT (HKCR)	File associations and COM information
HKEY_USERS (HKU)	Configuration for loaded user profiles
HKEY_CURRENT_CONFIG (HKCC)	Current hardware profile

Process Monitor is a Windows monitoring tool that records real-time activity involving:

- Registry
- File System
- Processes and Threads
- Network activity

For this experiment, we mainly analyze **Registry operations** and **boot-time activity**.

Part A — Registry Analysis

Step 1: Download Process Monitor

1. Open the Microsoft Sysinternals Process Monitor page.
2. Download the ZIP file.
3. Extract the ZIP file.
4. Open the extracted folder.
5. Run:
Procmon64.exe
6. If Windows displays a UAC prompt, select **Yes**.

Step 2: Start Process Monitor

When Procmon starts, it immediately begins collecting system events.
You may see thousands of events.

Press:

Ctrl + X

to clear the existing captured events.

Step 3: Configure Registry Filter

Open:

Filter → Filter...

Create the following filter:

Operation begins with Reg

Select:

Include

Then click **Add → Apply → OK**.

Now Procmon primarily displays Registry-related operations.

Step 4: Observe Registry Operations

You may see operations such as:

RegOpenKey

RegQueryValue

RegSetValue

RegCreateKey

RegDeleteKey

Observe the following columns:

Column	Description
Time of Day	Time when the event occurred
Process Name	Process generating the event
PID	Process ID
Operation	Registry operation
Path	Registry key/value accessed
Result	Success/failure result
Detail	Additional information

Step 5: Analyze an Event

Select any Registry event.

For example:

Process Name: explorer.exe

Operation: RegQueryValue

Result: **SUCCESS**

Double-click the event.

Examine:

- Registry path
- Operation
- Result
- Query information
- Process information
- Stack information

Record a few representative events in your lab observation table.

5. Registry Observation Table

You can use this format in your record:

S.No	Process	Operation	Registry Path	Result
1	explorer.exe	RegOpenKey	HKCU...	SUCCESS
2	svchost.exe	RegQueryValue	HKLM...	SUCCESS
3	RuntimeBroker.exe	RegQueryValue	HKCU...	SUCCESS
4	System process	RegOpenKey	HKLM...	NAME NOT FOUND

The actual values will depend on your Windows system and should be recorded from your own Procmon capture.

Part B — Boot-Time Logging

Step 6: Enable Boot Logging

In Process Monitor, select:

Options → Enable Boot Logging

A dialog box will appear.

Select the default boot logging configuration and click **OK**.

Step 7: Restart the Computer

Save your work and restart Windows.

During the next boot, Process Monitor records system activity occurring during startup.

This can include activity from:

- Windows services
- Startup applications
- Drivers
- System processes
- Registry operations
- File-system operations

Step 8: Open Process Monitor After Login

After Windows starts:

1. Log in normally.
2. Open **Process Monitor as Administrator**.
3. Procmon will detect the boot-time capture.
4. If prompted to save the boot log, choose **Yes**.
5. Save the file with a name such as:

Boot_Registry_Analysis.pml

Part C — Analyze Boot Activity

Step 9: Filter Registry Operations

Open:

Filter → **Filter...**

Add:

Operation begins with Reg

Select:

Include

Click:

Add → **Apply** → **OK**

Now you can examine Registry activity that occurred during boot.

Step 10: Analyze Failed Registry Operations

You can also look for unsuccessful operations.

For example, create a filter:

Result is NAME NOT FOUND

Select:

Include

This can reveal Registry paths that processes attempted to access but that were not present.

Important: NAME NOT FOUND is not automatically evidence of malware. Windows applications commonly check whether optional Registry keys or values exist.

Step 11: Identify Startup Processes

Examine the **Process Name** column.

Common processes may include:

System

services.exe

svchost.exe

explorer.exe

winlogon.exe

lsass.exe

Record unusual or unexpected processes separately and investigate their executable path and digital signature rather than assuming they are malicious.

6. Analysis Table

Prepare a table like this:

S.No	Time	Process	Operation	Registry Path	Result	Observation
1	08:30:01	services.exe	RegOpenKey	HKLM..	SUCCESS	Service configuration accessed
2	08:30:02	svchost.exe	RegQueryValue	HKLM..	SUCCESS	Configuration queried
3	08:30:04	explorer.exe	RegOpenKey	HKCU...	SUCCESS	User configuration accessed
4	08:30:05	application.exe	RegQueryValue	HKCU...	NAME NOT FOUND	Requested value unavailable

Use **your own captured events** for the final record.

7. Important Procmon Features Used

Filtering

Filtering reduces thousands of events to the events relevant to the investigation.

Example:

Operation begins with Reg

Event Properties

Double-clicking an event provides detailed information about the event.

Process Tree

Use:

Tools → Process Tree

to understand processes involved during startup.

Boot Logging

Options → Enable Boot Logging

allows Procmon to capture activity during Windows startup.

Save Capture

Use:

File → Save

and save the capture as:

.pml

8. Result

Registry activity and boot-time system activity were successfully captured and analyzed using Microsoft Sysinternals Process Monitor. Registry operations, processes, accessed Registry paths, and operation results were identified from the captured events.

Viva Questions

1. What is Process Monitor?

Process Monitor is a Windows Sysinternals tool used to monitor real-time file-system, Registry, process/thread, and related system activity.

2. What is Windows Registry?

It is a hierarchical database containing configuration information used by Windows and installed applications.

3. What is boot logging?

Boot logging captures system activity that occurs during the Windows startup process.

4. What does RegQueryValue mean?

It represents a process querying a Registry value.

5. What does NAME NOT FOUND indicate?

The requested Registry key/value was not found. It does **not by itself indicate malicious activity**.

6. What is a .PML file?

It is Process Monitor's native capture file format.

7. Why are filters used in Procmon?

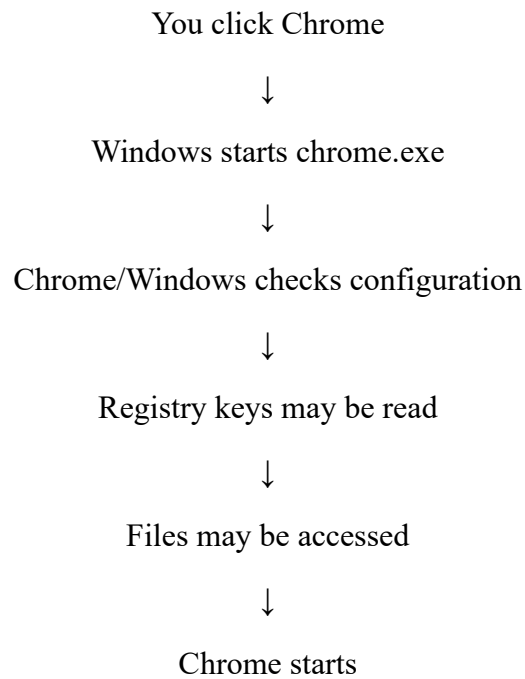
To reduce the large number of captured events and focus on events relevant to the investigation.

1. What are we actually doing?

Before opening Procmon, understand the basic idea.

When you use Windows, many things happen in the background.

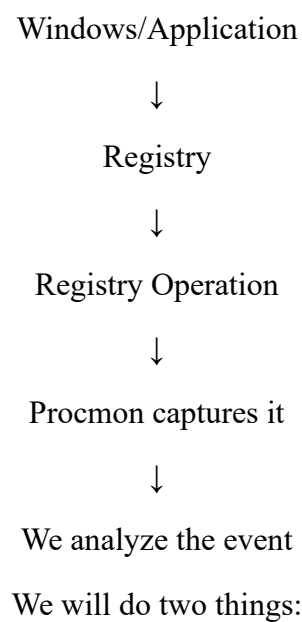
For example, when you open Chrome:



You normally don't see these operations.

Process Monitor (Procmon) allows us to see many of these hidden activities.

For this experiment, we are interested mainly in:



Experiment 1

Monitor **Registry activity** while Windows is running.

Experiment 2

Enable **Boot Logging**, restart Windows, and analyze Registry activity that happened during startup.

2. What is the Windows Registry?

Think of the Registry as a large configuration database used by Windows.

It stores information such as:

- Windows configuration
- User configuration
- Application settings
- Startup configuration
- Device configuration
- File associations
- Service configuration

For example, Windows may need to know:

"What program should open .pdf files?"

Some of this configuration is stored in the Registry.

3. What does a Registry path look like?

You may see something like:

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion

It looks complicated initially, but it is similar to folders.

For example:

HKEY_LOCAL_MACHINE

```
|
+-- Software
    |
    +-- Microsoft
        |
        +-- Windows
            |
            +-- CurrentVersion
```

There are different major Registry sections called **hives**.

The important ones for this experiment are:

Short name	Full name	General purpose
HKLM	HKEY_LOCAL_MACHINE	System-wide settings
HKCU	HKEY_CURRENT_USER	Current user's settings
HKCR	HKEY_CLASSES_ROOT	File associations/COM
HKU	HKEY_USERS	User profiles
HKCC	HKEY_CURRENT_CONFIG	Current hardware configuration

You don't need to memorize all of them today.

For this experiment, remember:

HKLM = computer/system configuration

HKCU = current user's configuration

4. What is Process Monitor?

Process Monitor is commonly called **Procmon**.

It is a Microsoft Sysinternals tool.

It can show activity involving:

- Registry
- Files
- Processes
- Threads
- Other system activity

Official Microsoft page:

[Microsoft Sysinternals Process Monitor](#)

5. Understanding the Procmon screen

When you first open Procmon, you may see thousands or even millions of events.

Don't panic. That's normal.

You will see columns similar to:

Time of Day

Process Name

PID

Operation

Path

Result

Detail

Let's understand each one.

Time of Day

Example:

10:35:21.1234567

This tells you **when the event occurred**.

Process Name

Example:

explorer.exe

svchost.exe

services.exe

chrome.exe

This tells you which process generated the event.

For example:

chrome.exe

means Chrome generated the event.

PID

PID means:

Process ID

Example:

PID: 12340

Windows assigns a numerical identifier to each running process.

You don't need to focus heavily on PID for your first experiment.

6. What is "Operation"?

This is one of the most important columns.

You may see:

RegOpenKey

RegQueryValue

RegSetValue

RegCreateKey

RegDeleteKey

These tell you what the process is doing with the Registry.

RegOpenKey

A program is opening/accessing a Registry key.

Example:

RegOpenKey

Think:

"I want to access this Registry location."

RegQueryValue

A program is reading a Registry value.

Think:

"What is stored here?"

RegSetValue

A program is writing/changing a Registry value.

Think:

"Change this configuration."

This operation deserves more attention during security analysis because something is **writing** to the Registry.

However, legitimate Windows applications also use it, so it isn't automatically suspicious.

RegCreateKey

A program creates a Registry key.

RegDeleteKey

A Registry key is deleted.

7. What is "Path"?

Example:

HKCU\Software\Microsoft\Windows\CurrentVersion\...

This tells you **which Registry location was accessed**.

This is very important when performing Registry analysis.

8. What is "Result"?

You might see:

SUCCESS

or

NAME NOT FOUND

or other results.

SUCCESS

The operation worked.

Example:

RegQueryValue → SUCCESS

means the process successfully queried the Registry value.

NAME NOT FOUND

The requested Registry key/value wasn't found.

Beginners often make this mistake:

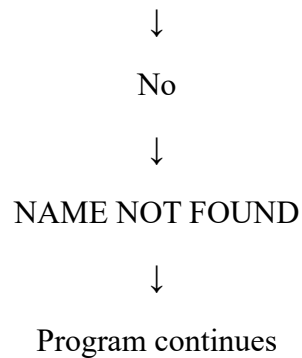
"NAME NOT FOUND = malware."

No.

That's incorrect.

Programs frequently check whether something exists:

Does this Registry value exist?



So we need context before deciding that something is suspicious.

9. First Practical Exercise — Registry Monitoring

Now let's actually perform the experiment.

Step 1 — Download Procmon

Download Process Monitor from Microsoft's Sysinternals page.

Extract the downloaded ZIP file.

You should find something similar to:

Procmon.exe

Procmon64.exe

Procmon64a.exe

On a normal modern 64-bit Windows computer, use:

Procmon64.exe

10. Step 2 — Run Procmon

Right-click:

Procmon64.exe

and select:

Run as administrator

Windows may show:

User Account Control

Do you want to allow this app...

Select:

Yes

11. Step 3 — First thing you will notice

Procmon may immediately start displaying events.

For example:

explorer.exe

svchost.exe

SearchHost.exe

RuntimeBroker.exe

...

There may be thousands of entries.

That's completely normal.

12. Step 4 — Stop the capture temporarily

Press:

Ctrl + E

This toggles event capturing.

When capture is stopped, you can examine the existing events without the list constantly changing.

Then clear the events:

Ctrl + X

Now the display should become empty.

13. Step 5 — Start capturing again

Press:

Ctrl + E

Now Procmon is recording new activity.

14. Step 6 — Create a Registry filter

This is the most important step.

Go to:

Filter → **Filter...**

You will see a filter window.

Select:

Column:

Operation

Condition:

begins with

Value:

Reg

Then choose:

Include

Click:

Add

Then:

Apply

Then:

OK

15. Why did we use "begins with Reg"?

Because Registry operations generally appear as:

RegOpenKey

RegQueryValue

RegSetValue

RegCreateKey

RegDeleteKey

They all begin with:

Reg

Therefore:

Operation begins with Reg

helps us focus on Registry operations.

16. Now generate some activity

Don't just stare at Procmon.

Perform normal actions on your computer.

For example:

1. Open File Explorer.
2. Open Settings.
3. Open Notepad.
4. Close Notepad.
5. Open your browser.
6. Close the browser.

Procmon should capture Registry operations generated by these applications.

17. Your screen might look like this

You may see something conceptually similar to:

Process Name	Operation	Path	Result
explorer.exe	RegOpenKey	HKCU\Software\...	SUCCESS
explorer.exe	RegQueryValue	HKCU\Software\...	SUCCESS
svchost.exe	RegOpenKey	HKLM\Software\...	SUCCESS

application.exe RegQueryValue HKCU\Software\... NAME NOT FOUND

Your actual values will be different.

Don't try to reproduce my exact paths.

Your job is to analyze the events generated on **your own computer**.

18. How to analyze one event

Choose one event.

For example:

explorer.exe

RegQueryValue

HKCU\Software\...

SUCCESS

Double-click the event.

A detailed window will appear.

Look at:

Event

You may see:

Operation: RegQueryValue

Result: SUCCESS

Process

You may see:

Process Name: explorer.exe

PID: xxxx

Stack

You may also see information about the execution stack.

For this beginner experiment, you don't need to understand every Stack entry.

19. Your first analysis question

For every event ask yourself:

Question 1

Which process performed the operation?

Example:

explorer.exe

Question 2

What operation did it perform?

Example:

RegQueryValue

Question 3

Which Registry path did it access?

Example:

HKCU\Software\...

Question 4

Did the operation succeed?

Example:

SUCCESS

That's basic Registry analysis.

20. Create your observation table

In your laboratory record, create:

S.No	Process	Operation	Registry Path	Result	Interpretation
1	explorer.exe	RegOpenKey	HKCU...	SUCCESS	Registry key accessed
2	explorer.exe	RegQueryValue	HKCU...	SUCCESS	Registry value read
3	svchost.exe	RegQueryValue	HKLM...	SUCCESS	System configuration queried

4	application.exe	RegQueryValue	HKCU...	NAME NOT FOUND	Requested value was absent
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Again, **replace these examples with your actual Procmon observations.**

21. Second Exercise — Boot-Time Logging

Now we move to the more interesting part.

We want to know:

What happens when Windows starts?

Normally you don't see all of this.

Boot logging allows Procmon to capture activity during the startup process.

22. Before enabling Boot Logging

Save your work.

Close applications you don't need.

Then open:

Process Monitor → Options

Look for:

Enable Boot Logging

Click:

Enable Boot Logging

Procmon will ask whether you want to enable boot logging.

Confirm it.

23. Restart Windows

Restart your computer normally.

For example:

Start → Power → Restart

Don't worry if you don't see Procmon during the boot process.

That's expected.

Procmon records the startup activity.

24. Log into Windows

After Windows starts:

1. Log into your account.
2. Wait until the desktop has loaded.
3. Start Procmon again if it isn't already open.

Procmon should detect the boot capture.

You may get a message asking whether you want to save the captured events.

Save it.

For example:

Boot_Logging_Experiment.pml

25. What is a PML file?

.PML is the native Process Monitor log format.

Think of it as:

Your investigation

↓

Procmon capture

↓

.PML file

You can reopen the file later using Process Monitor.

26. Analyze the boot log

Again go to:

Filter → Filter...

Use:

Operation begins with Reg

Select:

Include

Then:

Apply → OK

Now you're looking specifically at Registry activity during the captured boot process.

27. What should you look for?

Look at:

1. Which processes access Registry keys?

For example:

services.exe

svchost.exe

explorer.exe

2. Which Registry areas are accessed?

For example:

HKLM\Software\...

HKCU\Software\...

3. What operations are performed?

For example:

RegOpenKey

RegQueryValue

RegSetValue

4. Are there errors?

Look at:

NAME NOT FOUND

ACCESS DENIED

But remember:

An error does not automatically mean malicious activity.

28. A useful beginner investigation

Let's say you see:

Process: example.exe

Operation: RegQueryValue

Path:

HKCU\Software\Example\Application

Result:

SUCCESS

Your interpretation could be:

The process example.exe successfully queried a Registry value under the current user's configuration.

That's a factual observation.

Don't write:

"example.exe is malware."

You don't have enough evidence for that conclusion.

29. Another example

Suppose you see:

Process:

example.exe

Operation:

RegSetValue

Path:

HKCU\Software\Example\Settings

Result:

SUCCESS

Your observation:

example.exe successfully modified a Registry value.

That is interesting because it is a **write operation**.

Then you would investigate further:

- What is the application?
- Where is its executable located?
- Is it digitally signed?
- Is the Registry modification expected?
- Did the modification occur during installation/configuration?
- Is it associated with startup behavior?

This is how cybersecurity analysis should be performed.